NIRANJAN K R

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Professional Summary :

- A motivated B.Tech Biotechnology student with a strong interest in programming, seeking to learn, grow, and contribute through both biological and computational skills.
- To build a career that combines biotechnology and coding, by applying programming skills to solve real-world scientific and technical problems.
- To secure an opportunity where I can apply my knowledge of biotechnology along with my growing interest in coding, and continuously develop my technical and analytical skills.

Internships

Thangam Info Tech, Coimbatore

Full-Stack Development

February 2026 – April 2026

- Gained hands-on experience in web development using JavaScript, React, Tailwind CSS, and SQL Server, contributing to live web pages and UI development.
- Developed a Clinical Management System using React, TypeScript, Tailwind CSS, .NET Core Web API, and SQL Server to manage patient and appointment information.

Sree Mahalakshmi Technology Solution, Coimbatore

March 2026

- Gained hands-on experience in AI and Machine Learning using Python, dataset preprocessing, model development, and performance evaluation.
- Developed machine learning models using Python and the UCI Drug Consumption dataset to predict drug consumption patterns using classification algorithms.

BioXplora, Coimbatore

October 2025

- Gained knowledge of molecular docking, biological databases, and docking software for protein–ligand interaction analysis in drug discovery.
- Studied malaria by exploring potential drug targets and analyzing protein–ligand interactions using molecular docking techniques.

Biozone Research Technologies, Chennai

June 2026

- Gained hands-on experience in genomic DNA isolation, PCR, RAPD, RFLP, gel electrophoresis, DNA quantification, primer design, and bioinformatics tools such as NEBcutter and MEGA for molecular and sequence analysis.

Sri Shakthi Food Testing Laboratory, Coimbatore

September 2025

- Gained hands-on experience in food, water, and soil analysis through microbiological testing, chemical analysis, laboratory instruments, and titration methods while following food safety standards and laboratory SOPs.

V J Biotech, Coimbatore

January 2026

- Gained hands-on experience in chromatographic separation techniques, nanoparticle synthesis, biological activity testing, and bio-pharmaceutical and natural product studies.

Projects

Full-Stack Application - Clinical Management System

- Developed a clinical management web application using React, TypeScript, Tailwind CSS, .NET Core Web API, and SQL Server to manage patient and appointment information.
- Implemented user interfaces and API integration for patient records, appointment scheduling, and clinical data management.

Drug Consumption Prediction using Machine Learning

- Developed a machine learning model using Python and demographic and psychological data from the UCI Drug Consumption dataset to analyze drug consumption patterns.
- Implemented and compared Logistic Regression, Decision Tree, and Random Forest algorithms to evaluate model performance and prediction accuracy.

Microbial Diversity Analysis of Pork Intramuscular Tissue:

- Identified and characterized bacterial diversity using microbial culturing and 16S rRNA gene sequencing, with sequence analysis and NCBI database submission.

Phytochemical Analysis and Antidiabetic Potential of Syzygium cumini:

- Evaluated seed phytochemicals and assessed antidiabetic, antimicrobial, and antioxidant activities using standard laboratory assays.

Dye Degradation Using Honey-Mediated Silver Nanoparticles:

- Synthesized silver nanoparticles using natural honey and evaluated their photocatalytic efficiency in degrading methylene blue dye under sunlight.

Isolation of Oil-Degrading Bacteria from Petroleum-Contaminated Soil:

- Isolated and screened oil-degrading bacteria from contaminated soil to investigate their potential for bioremediation.

Education

Bachelor of Technology (B.Tech.) – Biotechnology (2024 – 2028)

CGPA- 8.13

Sri Shakthi Institute Of Engineering And Technolology, Coimbatore

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Key Skills

Frontend : React • TypeScript • JavaScript • HTML • CSS • Tailwind CSS • Responsive UI • Lifecycle Hooks

Backend & APIs : .NET Core Web API • Node JS • API Integration

Database : SQL Server • MySQL • UCI Repo

Tools & Methodologies : Git • Git Version Control (branching, merging, pull requests) • GitHub • VS Code • Visual Studio

AI Skills : Model Development • Algorithms Implementation • AI tools

Programming : C • Python

Biotecnology & Lab : Bioinformatics • Wet lab • Experimental Biotechnology • Analytical techniques

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Achievements

- **NCBI GenBank Sequence Publication :** Published a 16S rRNA gene sequence in NCBI GenBank (Accession No.: PQ772087.1). Submitted multiple prokaryotic 16S rRNA gene sequences for public database publication.
- **Research Paper Presentation - ICFBB 2025 :** Presented research on phytochemical analysis & alpha-glucosidase inhibition for diabetes management
- **Dr.Kalam Young Achiever Award 2026 :** Recognized for an engineering project demonstrating engineering aptitude.

Certifications

- **AI Tools Workshop:** Completed a workshop by Ben10x on AI-assisted presentations, data analysis, coding & debugging.
- **AI Webinar :** Participated in a two-day AI webinar conducted by AI Activation Academy.
- **Plant Developmental Biology :** NPTEL elite certification from IIT Roorkee with 68% score.
- **ARMS-PCR Workshop :** Participated in the “Decoding Genetic Variation - ARMS PCR” workshop.
- **Mushroom Spawn Cultivation :** Completed a three-day online workshop conducted by Flavin Labs Pvt. Ltd.
